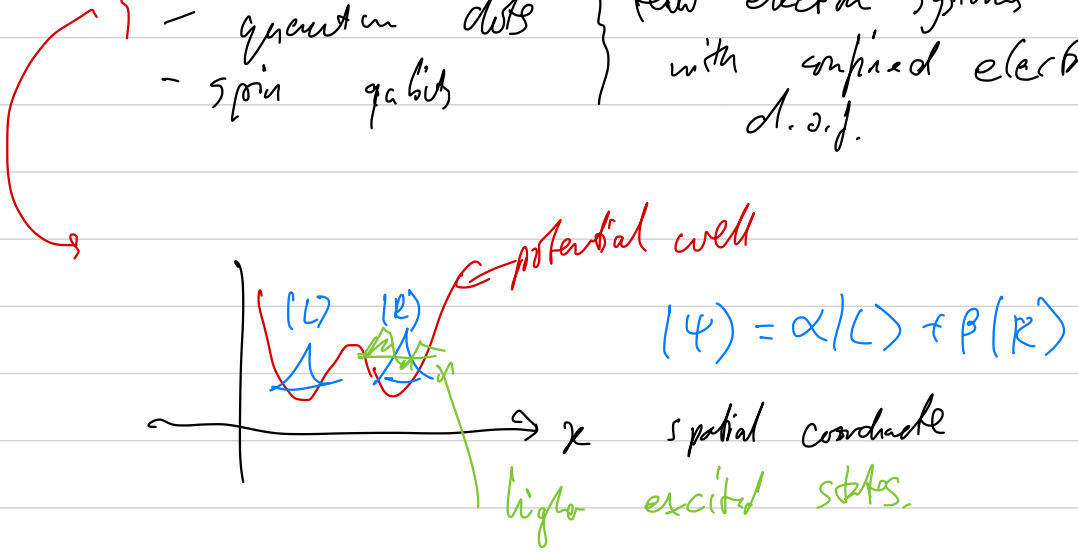
2026

Solid State Qubits

Different types: but all based on "condensed matter" systems, electronic d.o.f. → qubit

- Semiconductor devices (next week)
  - charge-qubits
  - quantum dots
  - spin qubits

  } few-electron systems with confined electronic d.o.f.



- "artificial atom"

- Gallium Arsenide – Indium Arsenide
- Nitrogen vacancies in diamond
- CMOS to build quantum dots

- Ion-traps (matter-based)

Superconducting Qubits

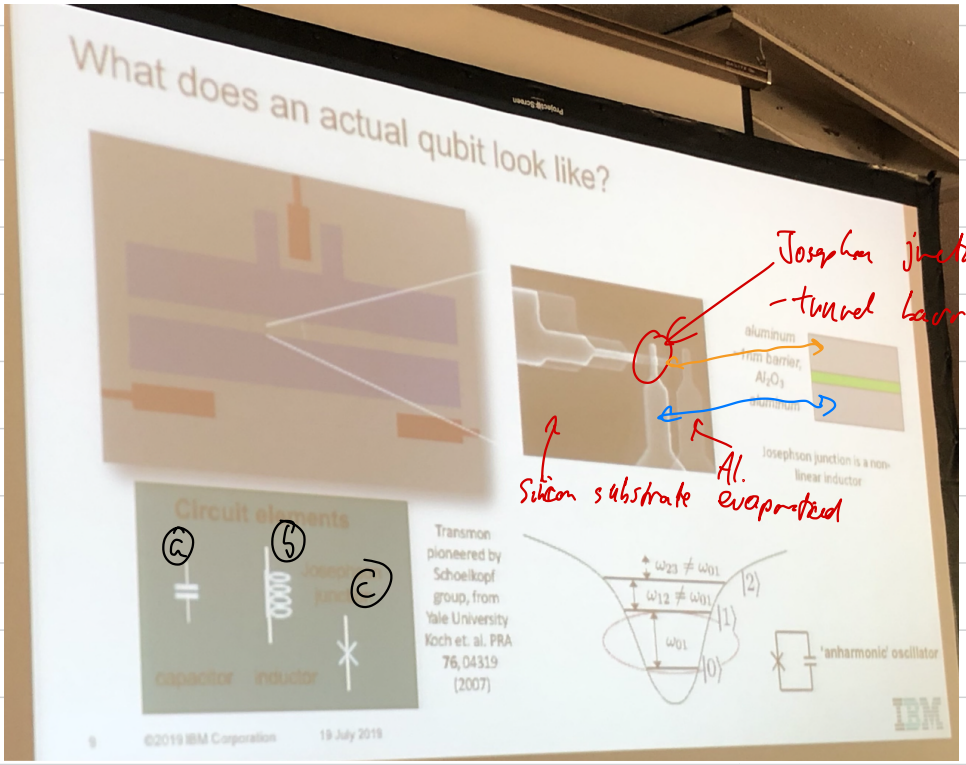
- Google, IBM, Rigetti, Intel (last year)
- encoding q. info in superconducting electronics
- Superconductivity is "dissipationless"
  - this week: quantized circuit theory
- "Many-body" systems:
  - electrons can have an attractive potential in a lattice
    ⇒ bind into "Cooper pairs"
  - ground state of a superconducting metal is called the "BCS" state
    Bardeen, Cooper & Schrieffer.
    [won a Nobel prize]
- 2025 Nobel prize was on this topic
- Zoology of superconducting qubits
  - "charge qubits"/"Cooper-pair box" (transmons"
  - "phase qubits"
  - "flux qubits" & "fluxonium"
  - "0-π" qubits
  - "dualon"

  (different electronic circuits)

⇒ energy e/states ⇒ "good" qubits.

Goal: understand basic idea of circuit quantisation
→ "LC" oscillator
→ Cooper-pair box / transmon

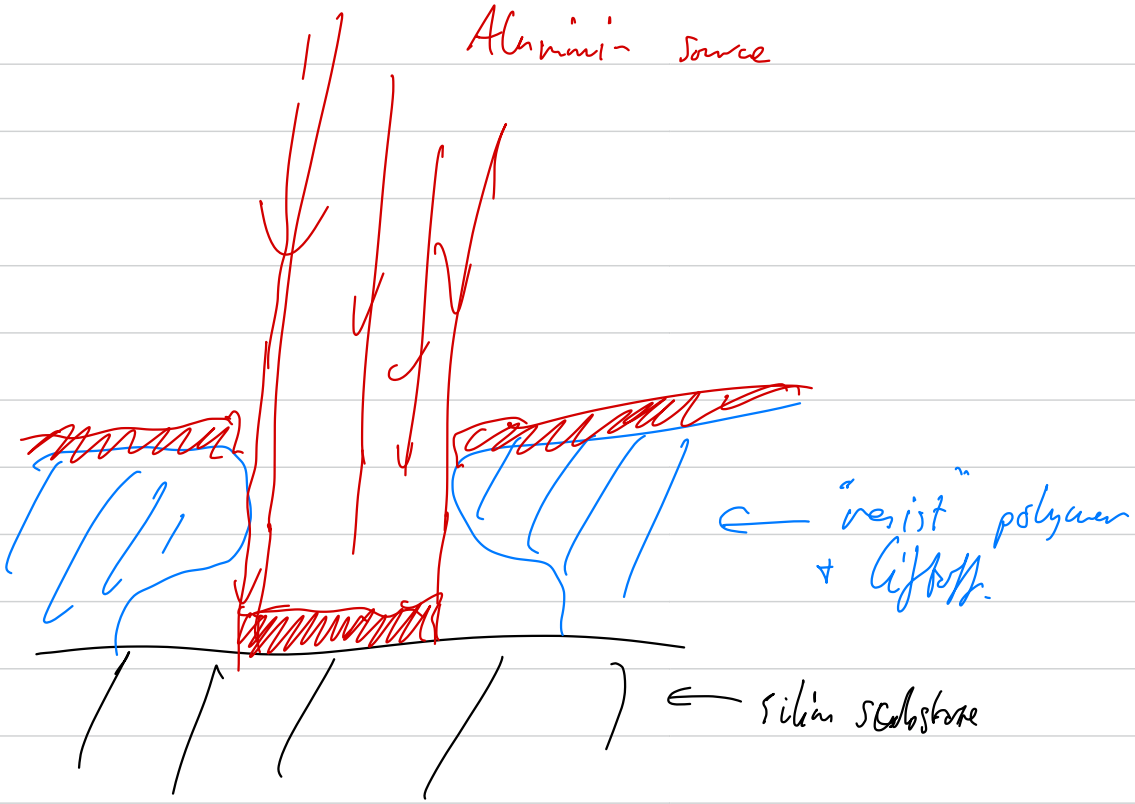
Plan: revise circuit theory
— E.o.m for electrical circuits
→ Lagrangian / Hamiltonians
→ canonical variables (analogues to x + p)
→ Canonical quantisation
→ put circuits elements together to make a qubit.



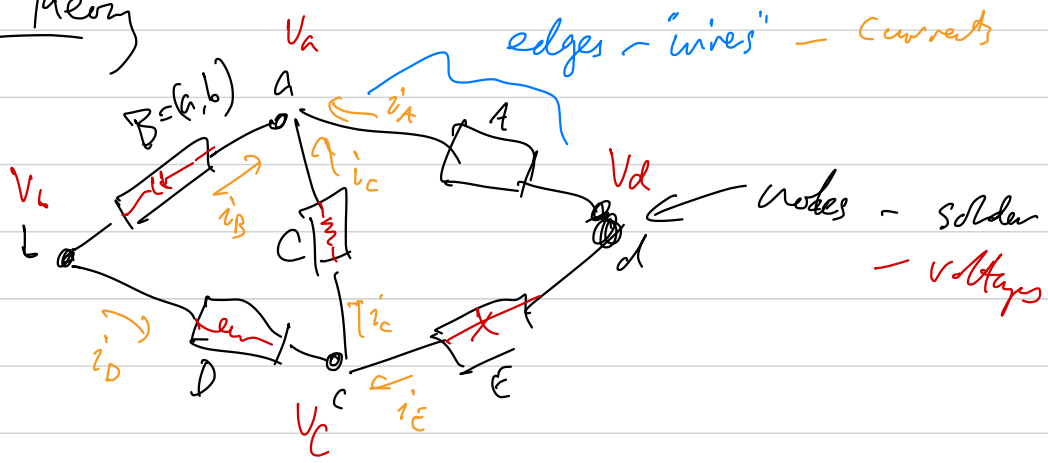
Superconducting qubits are built from ⓐ Capacitors, ⓑ inductors & ⓒ josephson junctions

- operate at ultra-cold temperatures
  - superconducting transition temp. ~ 1 – 10 K
  - superconducting qubits ~ 0.01 K.
  - outer space is @ 2.7 K

fabrication method: aluminum evaporation



Circuit Theory



Graph :

for quantum superconducting systems → no resistors

- Task : derive the E.O.M for a circuit "graph"
  - dynamics of the circuit coordinates / degrees of freedom
  - conservation laws (energy, mass, charge)
    - accounting rules

(1) – Kirchoff's voltage law (conservation of energy around a loop in the circuit)

$$\sum_{\text{Loop } \ell} \text{Voltage rises around the loop} = 0$$

eg $$V_{ac} + V_{cb} + V_{ba} = 0$$

- where $V_{ac} = V_c - V_a$, $V_{cb} = V_b - V_c$, $V_{ba} = V_a - V_b$

"solves" the loop eqn around Loop $a \to c \to b \to a$

or $$\boxed{V_C + V_D + V_B = 0}$$ KVL for a Loop

(also $V_A + V_B + V_C = 0$ for the circuit above)

② Kirchoff's Current Law (conservation of charge)

$$\sum_{\substack{\text{Currents along edges attached to a node}}} (\text{Current into node})_{\text{edge}} = 0$$

eg $i_A + i_B + i_C = 0$ for node a

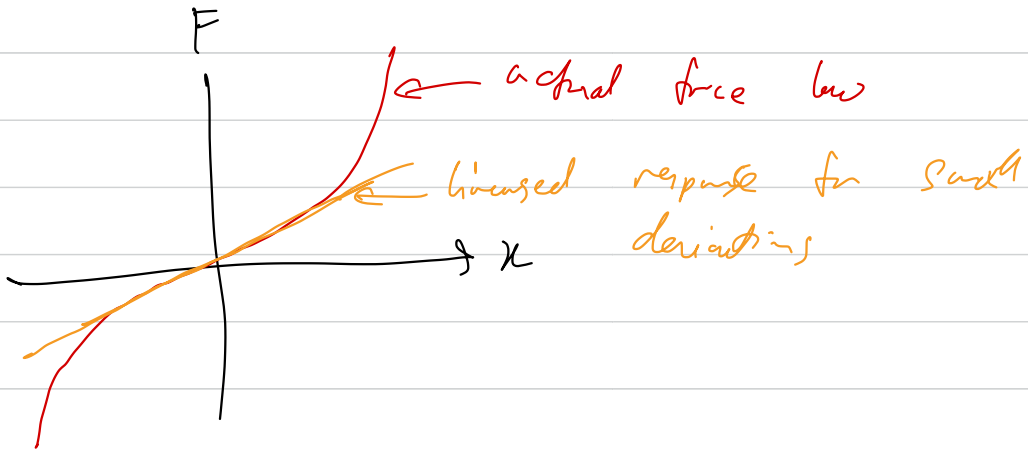
$i_D + i_E - i_C = 0$ for node c

$-i_B - i_D = 0$ for " node b

$-i_A - i_E = 0$ " d

③ Constitutive Laws
- specific for each element
- parameterised by element property (eg capacitance, inductance, resistance)
- depend on the material, geometry etc
- approximate
- a bit like pressure × volume of a gas

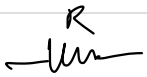
mechanical analogue: Spring — (ideal) linear spring $F = -kx$
— nonlinear " $F = F(x)$
$= -kx + \alpha x^2 + \dots$

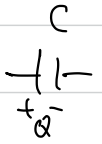


- also $\underline{F} = m\underline{a}$

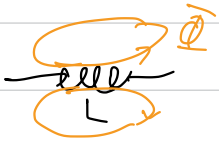
$\quad = m(x, v)\,\underline{a}$

For electrical systems relate voltage / current to flux or charge.

resistors  $V = i \cdot R$

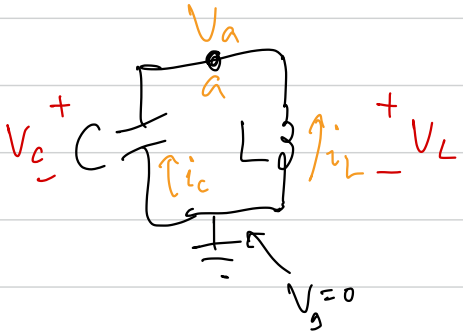
Capacitors  $V = Q/C$

$\Rightarrow \dfrac{dV}{dt} = \dfrac{1}{C}\dfrac{dQ}{dt} = \dfrac{i}{C}$

inductor  $V = -L\dfrac{di}{dt}$

define $V = \dot{\Phi} \Rightarrow \Phi = -Li$

Electrical Simple Harmonic Oscillator



① KVL around the loop

$V_c - V_L = (V_a - V_g) - (V_a - V_g) = 0$

$\Rightarrow V_c = V_L = V_a$

(2) KCL @ node a $\quad i_C + i_L = 0$

$\Rightarrow \quad i_C = -i_L = i$ ← Loop current

(3) constitutive laws

Capacitance (Farads) · Charge (Coulombs)

$C V_C = Q_C \Rightarrow C \frac{dV_C}{dt} = \frac{dQ_C}{dt} = i_C = i$ (3a)

(V_C: Volts)

$\Phi_L = -iL \Rightarrow V_L = -L \frac{di}{dt}$ (3b)

flux (Tesla) · current (amps) · inductance (Henries) · $V_L = \dot{\Phi}_L$

(2) + (3a) $\Rightarrow \quad i = i_C = -i_L = C \frac{dV_a}{dt}$

(1) + (3b) $\Rightarrow \quad V_a = V_C = V_L = -L \frac{di}{dt}$

} 2 eqns for 2 unknowns i, V_a

$\Rightarrow \quad V_a = -L \frac{di}{dt} = -L \frac{d}{dt}\left(C \frac{dV_a}{dt} \right) = -LC \frac{d^2 V_a}{dt^2}$

$\Rightarrow \quad \boxed{LC \ddot{V}_a + V_a = 0}$ (4) — 2nd order ODE for V_a

+ $\quad i = C \frac{dV_a}{dt} = C \frac{d}{dt}\left(-L \frac{di}{dt} \right) = -LC \frac{d^2 i}{dt^2}$

$\Rightarrow \quad LC \ddot{i} + i = 0$ — 2nd order ODE for i

we have the LC-circuit E.O.M

$\Rightarrow$ 2nd order ODE

$\Rightarrow$ 2 initial conditions to solve for V or i

Solution for (1)

$$LC\ddot{V}_a + V_a = 0 \quad (2)$$

try: $V_a(t) = A\sin(\omega t) + B\cos(\omega t)$

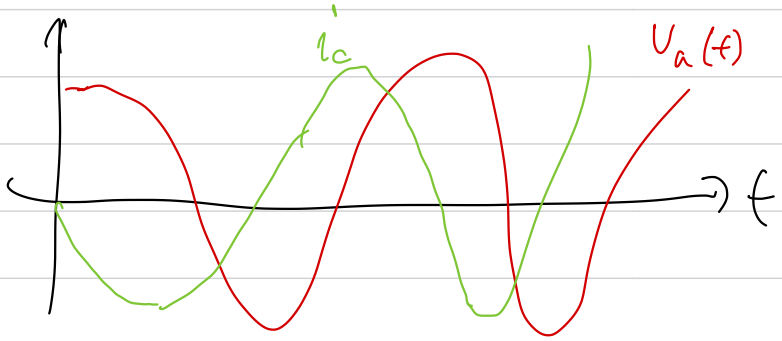
$$\Rightarrow LC\ddot{V}_a + V_a = LC(-\omega^2 A\sin\omega t - \omega^2 B\cos\omega t) + (A\sin\omega t + B\cos\omega t)$$

$$= A\sin\omega t\left(1 - LC\omega^2\right) + B\cos\omega t\left(1 - LC\omega^2\right)$$

$$= 0$$

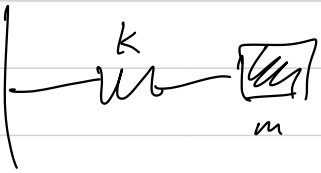
$$\Rightarrow LC\omega^2 = 1 \Rightarrow \omega = \frac{1}{\sqrt{LC}} = 2\pi f$$

so "LC" single harmonic oscillator has angular frequency solution with $\omega = \frac{1}{\sqrt{LC}}$



$$i_C = C\frac{dV_a}{dt}$$

Mechanical analogy



mass-spring system

E.O.M $F = -kx \Rightarrow ma = m\ddot{x}$

$\Rightarrow \frac{1}{C}V = -c\ddot{V}$

mech	⟷	electrical
x	⟷	V
velocity $= \dot{x}$	⟷	$\frac{dV}{dt}$
p	⟷	$i = C\frac{dV}{dt}$
mass	⟷	C
k	⟷	$\frac{1}{C}$
$\omega = \sqrt{\frac{k}{m}}$	⟷	$\frac{1}{\sqrt{LC}}$

$\frac{1}{C}i = -L\ddot{i}$

mech	⟷	electrical
x	⟷	i
vel $= \dot{x}$	⟷	$\frac{di}{dt}$
p	⟷	$V = L\frac{di}{dt}$
mass	⟷	L
k	⟷	$\frac{1}{C}$
$\omega = \sqrt{\frac{k}{m}}$	⟷	$\frac{1}{\sqrt{LC}}$

our "framework"

- The mechanical analogy will help to motivate how we quantise the system

→ "quantise" circuit coordinates just like a mechanical system.

→ "correspondence" principle : E.O.M → Lagrangian / Hamiltonian → quantum "

Circuit quantisation

Summary of Recipe

(0) derive classical E.O.M eg. $\ddot{x} = F(x)$

(1) find a Lagrangian $L(x, \dot{x})$ that generates E.O.M
- Euler-Lagrange equations of motion for $\ddot{x}(x)$

$$\frac{d}{dt}\frac{\partial L}{\partial \dot{x}} = \frac{\partial L}{\partial x}$$

(2) → Use $L(x, \dot{x})$ to compute conjugate momentum

$$p(\dot{x}) = \frac{\partial L}{\partial \dot{x}}$$

(3) Use L & p to construct the Hamiltonian $H(x, p)$
- Hamilton-Jacobi equations of motion for $\dot{x}$ & $\dot{p}$

$$H = p\dot{x} - L$$

(4) quantise $\hat{x}$, $\hat{p}$, $\hat{H}$ $\quad [\hat{x}, \hat{p}] = i\hbar$

9/10/2026

Mech. example : x is the position of a particle in a potential V

$$\mathcal{L} = \underbrace{T(\dot{x})}_{\frac{1}{2}m\dot{x}^2} - V(x)$$

E-L eqn : (1) $\frac{d}{dt}\frac{\partial \mathcal{L}}{\partial \dot{x}} = \frac{d}{dt}T(\dot{x}) = \frac{\partial \mathcal{L}}{\partial x} = -V'(x)$

$= \frac{d}{dt}(m\dot{x})$

$\Rightarrow \quad m\ddot{x} = -V'(x)$ which is $ma = F(x)$

(2) so $p = \frac{\partial \mathcal{L}}{\partial \dot{x}} = m\dot{x} = mv$

(3) & $H = p\dot{x} - \mathcal{L} = m\dot{x}^2 - \left(T(\dot{x}) - V(x)\right)$

$= \frac{1}{2}m\dot{x}^2 + V(x)$

but $\dot{x} = p/m$

$\Rightarrow \quad H(x,p) = \frac{p^2}{2m} + V(x)$

ex check Hamilton-Jacobi eqn $\Rightarrow$ $F=ma$ also

i.e. $\frac{\partial p}{\partial t} = -\frac{\partial H}{\partial x}$ $\quad \frac{\partial x}{\partial t} = \frac{\partial H}{\partial p}$

(4) quantise $H \rightsquigarrow \hat{H}$, $x \to \hat{x}$, $p \to \hat{p}$ & $[\hat{x}, \hat{p}] = i\hbar$

State space (x,p) $\longrightarrow$ Hilbert space $|\psi\rangle \in \mathcal{H}$

for a particle in a well

$$\mathcal{H} = \text{span}\{|x\rangle\}_x = \text{span}\{|p\rangle\}_p$$

$$\hat{x} = \int_{-\infty}^{\infty} dx \; x\,|x\rangle\langle x| \quad , \quad \hat{p} = \int_{-\infty}^{\infty} dp \; p\,|p\rangle\langle p|$$

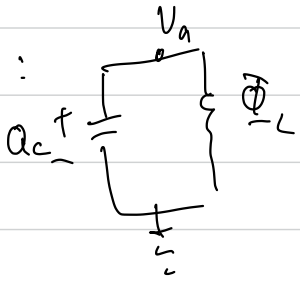
$$\hat{x}|x\rangle = x|x\rangle \qquad \hat{p}|p\rangle = p|p\rangle$$

Back to circuit theory

- on Wed. we wrote the circuit eqns
  KVL + KCL + constitutive eqns
  in terms of voltage V + current I
  which are derived quantities
- we reformulate in terms of
  underlying flux Φ + charge Q

$$V = \dot{\Phi} \qquad I = \dot{Q}$$

KVL:



$$\sum_{\text{loop edges}} V_{\text{rise edge}} = 0$$

$$\Rightarrow \dot{\Phi}_{cap} - \dot{\Phi}_L = 0$$

$$\Rightarrow \dot{\Phi}_{cap} - \dot{\Phi}_L = \dot{\Phi}_{const} \quad (= 0 \text{ for many cases})$$

KCL

$$\sum_{\substack{\text{edges} \\ \text{at node}}} i_{edge} = 0 = \sum \dot{Q}_{edge}$$

$$\dot{Q}_C + \dot{Q}_L = 0 \Rightarrow Q_C + Q_L = Q_{const}$$

Constitutive law

$$Q_C = C V_C = C \dot{\Phi}_C \Rightarrow Q_a = C \dot{\Phi}_a$$

$$\Phi_L = -L i_L = -L \dot{Q}_L \Rightarrow \Phi_a = -L \dot{Q}_a$$

$$\Rightarrow \Phi_a = -L \dot{Q}_a = -L (C \ddot{\Phi}_a) \quad (*)$$

$$\Rightarrow \Phi_a = A \cos \omega t + B \cos \omega t \qquad \omega = \frac{1}{\sqrt{LC}}$$

$$+ \quad Q_a = C \dot{\Phi}_A = -LC \ddot{Q}_a$$

(we could compute $V_a = \dot{\Phi}_a$, $i_a = \dot{Q}_a$)

- we take Φ_a as the circuit coordinate

so $$\mathcal{L}(\Phi, \dot{\Phi}) = T(\dot{\Phi}) - V(\Phi)$$

$$= \frac{C \dot{\Phi}^2}{2} - \frac{1}{2L} \Phi^2$$

check Euler Lagrange eqn:

$$\frac{d}{dt} \frac{\partial \mathcal{L}}{\partial \dot{\Phi}} = \frac{d}{dt} \left(C \dot{\Phi} \right) = C \ddot{\Phi}$$

$$= \frac{\partial \mathcal{L}}{\partial \Phi} = - \frac{\Phi}{L}$$

$$\Rightarrow \quad C \ddot{\Phi} = - \frac{\Phi}{L} \Rightarrow \quad LC \ddot{\Phi} = - \Phi = \text{(A)}$$ ✓

- Then conjugate "momenta" to the flux is

$$\frac{\partial \mathcal{L}}{\partial \dot{\Phi}} = C \dot{\Phi} = Q_a$$ from previous page

$$\Rightarrow \dot{\Phi} = \frac{Q_a}{C}$$

- Hamiltonian $H = \dot{\Phi} Q_a - \mathcal{L}(\Phi, \dot{\Phi})$

$$= \frac{Q_a^2}{C} - \left(T(\dot{\Phi}) - V(\Phi) \right)$$

$$= \frac{Q_a^2}{2C} + V(\Phi)$$

for a Harmonic oscillator LC circuit

$$H = \underbrace{\frac{Q^2}{2C}}_{\text{kinetic part, depending on "momentum"}} + \underbrace{\frac{\Phi^2}{2L}}_{\text{potential}} \quad \text{quantize} \Rightarrow \hat{H} = \frac{\hat{Q}^2}{2C} + \frac{\hat{\Phi}^2}{2L}$$

$$[\hat{\Phi}, \hat{Q}] = i\hbar$$

non-dimensionalise flux & charge

$$\hat{n} = \frac{\hat{Q}}{2e} \qquad \hat{\phi} = 2\pi \frac{\hat{\Phi}}{\Phi_0} \qquad \Phi_0 = \frac{h}{2e} \text{ is the "flux quantum"}$$

"charge number" "superconducting phase"

$$[\hat{n}, \hat{\phi}] = \left[\frac{\hat{Q}}{2e}, 2\pi\frac{\hat{\Phi}}{\Phi_0}\right] = \frac{2\pi}{2e\Phi_0}[\hat{Q}, \hat{\Phi}] = \frac{2\pi}{2e\Phi_0} i\hbar$$

$$= \frac{2\pi}{2e\left(\frac{h}{2e}\right)} i\hbar = i$$

$$\Rightarrow [\hat{n}, \hat{\phi}] = i$$

$$\hat{H} = \frac{\hat{Q}^2}{2C} + \frac{\hat{\Phi}^2}{2L} = \underbrace{\frac{(2e)^2}{2C}}_{E_C} \hat{n}^2 + \underbrace{\frac{\Phi_0^2}{8\pi^2 L}}_{E_L} \hat{\phi}^2$$

E_L ← dimension of Joules

$$= E_C \hat{n}^2 + E_L \hat{\phi}^2$$

Hilbert space operators

$$\hat{\phi} = \int_{-\infty}^{\infty} d\phi \;\; \phi \, |\phi\rangle\langle\phi| \qquad \hat{n} = \int_{-\infty}^{\infty} dn \; n \, |n\rangle\langle n|$$

$$\langle \phi | \hat{n} | \psi \rangle = -i \frac{\partial \psi}{\partial \phi}$$

[c.f. $\langle x | \hat{p} | \psi \rangle = -\frac{i}{\hbar} \frac{\partial \psi}{\partial x}$]

So we can "solve" the Hamiltonian in eg ϕ co-basis by evaluating the eval problem: $\hat{H}|\psi\rangle = E|\psi\rangle$

for the time-indep Schrödinger equation.

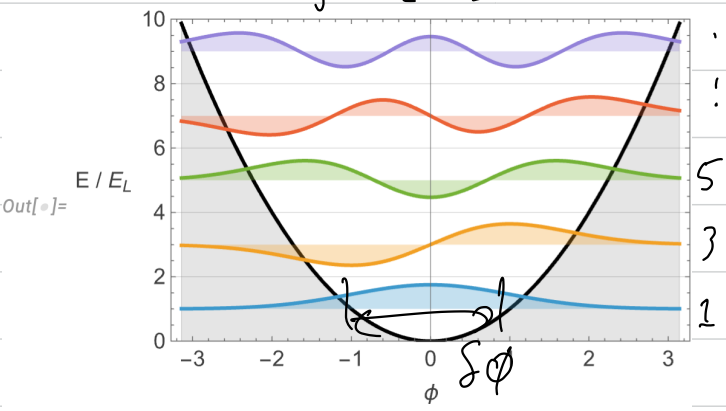
$$\Rightarrow \quad \langle \phi | \hat{H} | \psi \rangle = E \langle \phi | \psi \rangle \qquad \text{where } \hat{H} = E_c \hat{n}^2 + E_L \hat{\phi}^2$$

$$-E_c \frac{\partial^2 \psi}{\partial \phi^2} + E_L \phi^2 \psi(\phi) = E \psi(\phi) \qquad \text{where } \psi(\phi) = \langle \phi | \psi \rangle$$

$E_J = E_c = 1$

e/solutions

$\Rightarrow$



Out[ ]=

we learn from mech Harmonic osc.

$$E_j = 2\left(n + \frac{1}{2}\right)\omega_0 \qquad \omega_0 = \frac{1}{\sqrt{LC}} = \sqrt{E_J E_C}$$

$$\delta\phi = \sqrt{\frac{E_C}{E_L}} \quad \text{"zero point" fluctuations in } \phi$$

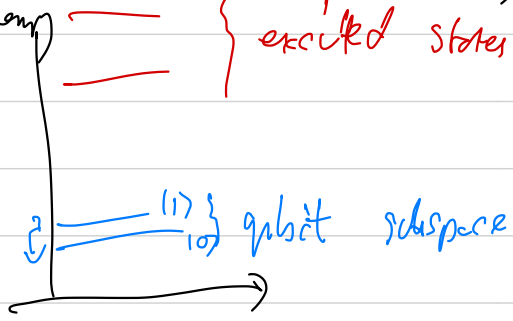
- So an LC electrical circuit has discrete energy e/states ✓
- energy spacing in "Harmonic" = equally spaced

bad for a qubit

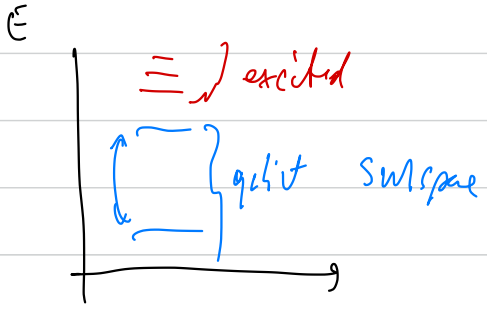
Since it is difficult to isolate just 2 e/states for the $|0\rangle, |1\rangle$ qubit basis

⇒ we need an anharmonic oscillator for a good electrical qubit.

Ideally • we want an isolated low energy qubit subspace
& the other states being very energetic



- we can make do with

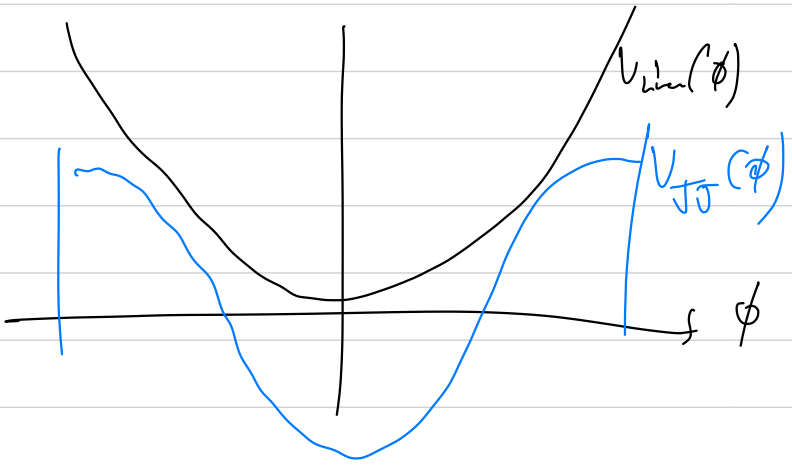


- As long as the energy spectrum is "anharmonic" we can (approx) isolate 2 states to be the qubit states

- A josephson junction is a nonlinear inductor:

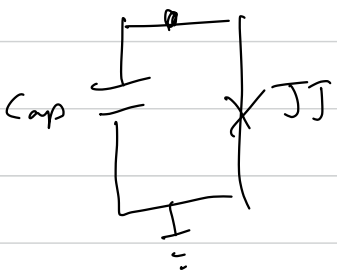
Linear inductor $V_{lin}(\phi) = E_L \phi^2$

non linear inductor $V_{JJ}(\phi) = -E_J \cos\phi$ ⊗

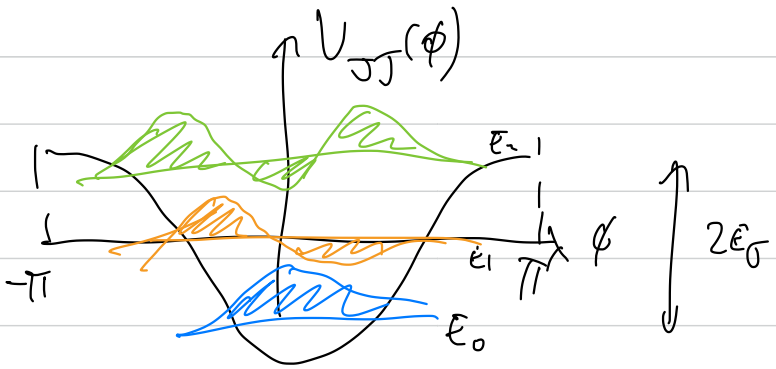


Josephson proved ⊗ in 1970s

⇒ circuit with a JJ instead of an inductor

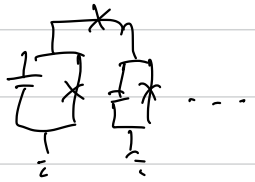


$$H = E_C \hat{n}^2 - E_J \cos(\hat{\phi})$$

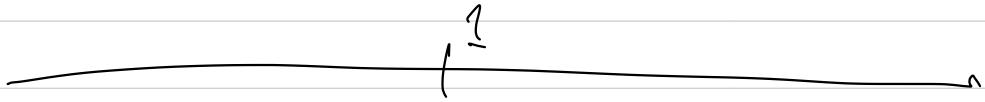


when $E_2 - E_1 \neq E_1 - E_0$ ⇒ anharmonic

So this is the basis for analysing superconducting circuits

- multiple qubits by copying  ...
  - coupled by other circuit elements

Tooling of S.C qubits



- Cooper-pair box
  1999
  - very anharmonic
  - noisy
- Transmon
  2007
  - quite harmonic
  - not very noisy

- Arrays of S.C. qubits being developed for quantum algorithms eg IBM, google